

Day 14 Hybrid Notes

Drawing right-triangle models for real situations | Cloze + outline + active learning

Name:	Date:	Period:
-------	-------	---------

Learning Targets
I can distinguish angles of elevation and depression.
I can draw and label a right-triangle model from a context.
I can use trigonometry to find and interpret heights and distances.

Part 1: Cloze Notes - Essential Ideas
1. An angle of elevation is measured upward from a horizontal _____.
2. An angle of depression is measured downward from a horizontal _____.
3. The line from the observer to the object is the line of _____.
4. Horizontal lines in an elevation/depression model are _____.
5. The angle of depression equals the alternate interior angle of _____.
6. Vertical height and horizontal distance are connected by _____.
7. In eye-level problems, add the observer's eye height to the triangle _____.

Part 2: Outline Notes - Big Picture A. A word problem becomes manageable after drawing a right-triangle _____. B. The reference angle must be measured from a horizontal _____. C. Side labels determine the correct trigonometric _____. D. A final answer needs reasonable rounding and correct _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ —
---	--

Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 14 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. A student stands 30 ft from a pole. The angle of elevation is 42 degrees. Set up the equation for height h .
Setup/work: _____
Answer: _____
2. From a 55-ft building, the angle of depression to a car is 28 degrees. Set up the equation for horizontal distance d .
Setup/work: _____
Answer: _____

Part 4: Error Analysis

A student reports only the triangle height when the observer's eye level is 5 ft above the ground.
What went wrong?

Correct idea:

—

Before You Solve Checklist

- ☐ I labeled the diagram or graph.
- ☐ I chose the correct relationship or formula.
- ☐ I substituted values carefully.
- ☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that _____.

Sentence 2: One thing I need to remember when solving is _____.

Part 5C: Application Problem

A surveyor's instrument is 4 ft high and 50 ft from a tree. The angle of elevation to the top is 39 degrees. Find the total tree height.
Setup: _____
Answer: _____
Sentence answer: _____